Cyberbullying Detection — Complete Comparison Tables

RoBERTa + LIME based approach vs. Base Paper and Related Work

1. Our Model Results (on CyberBullying Data.csv, ~4000 rows)

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression (C=1)	83.33%	83.62%	83.33%	83.27%
LSTM	87.33%	87.33%	87.33%	87.33%
GRU	85.33%	85.36%	85.33%	85.34%
RoBERTa (proposed)	**92.33%**	**92.36%**	**92.33%**	**92.33%**

RoBERTa Confusion Matrix (test n=600): TN=290, FP=19, FN=27, TP=264.

2. Comparison with Base Paper — Esan et al. (2024)

Format A — Same Model (RoBERTa), Different Dataset

Dataset	RoBERTa Accuracy
Our dataset (CyberBullying Data.csv, ~4,000 rows)	**92.33%**
Esan et al. dataset (46,017 rows, full)	89.01%

Format B — Same Dataset (Esan et al., 46,017 rows), Different Model

Model	Accuracy	Precision	Recall	F1-Score
Esan et al. LSTM (paper-reported)	77.00%	-	-	-
Our Logistic Regression (reproduced)	86.63%	85.24%	86.63%	85.12%
Our LSTM (reproduced)	86.82%	87.71%	86.82%	87.19%
Our GRU (reproduced)	87.12%	86.51%	87.12%	86.74%
Our RoBERTa (proposed)	**89.01%**	**88.95%**	**89.01%**	**88.98%**

Note: Our RoBERTa outperforms the base paper's own reported LSTM result (77.00%) by ~12 percentage points, evaluated on their own dataset.

3. Comparison with Ogunleye & Dharmaraj (2023)

Reference: Ogunleye, B., & Dharmaraj, B. (2023). "The Use of a Large Language Model for Cyberbullying Detection." Analytics, 2(3), 694-707. DOI: 10.3390/analytics2030038

3.1 Four-Way Experimental Comparison

Setting	Dataset	RoBERTa Accuracy	RoBERTa F1-Score
Our Dataset + Our Code	Our ~4,000 rows	92.33%	92.33%
Our Dataset + Their Code	**Our ~4,000 rows**	**95.25%**	**95.12%**
Their Dataset 3 + Our Code	38,204 rows (26,742/5,731/5,731 split)	89.79%	89.79%
Their Dataset + Their Published Results	Their D2 (~39,079 rows)	87.00%	87.00%

3.2 Full Model Breakdown — Our Dataset + Their (Ogunleye) Code

Model	Accuracy	Precision	Recall	F1-Score
TF-IDF + RF	83.25%	91.50%	72.16%	80.69%
TF-IDF + SVM	85.50%	88.64%	80.41%	84.32%
SBERT + RF	85.75%	87.43%	82.47%	84.88%
SBERT + SVM	87.00%	87.77%	85.05%	86.39%
XLNet	90.50%	91.05%	89.18%	90.10%
BERT	91.75%	92.59%	90.21%	91.38%
XLM-RoBERTa	93.00%	90.69%	95.36%	92.96%
RoBERTa	**95.25%**	**94.87%**	**95.36%**	**95.12%**

3.3 Full Model Breakdown — Their Dataset 3 + Our Code

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	86.91%	87.01%	86.91%	86.91%
GRU	86.58%	86.75%	86.58%	86.58%
LSTM	87.04%	87.27%	87.04%	87.03%
RoBERTa	**89.79%**	**89.79%**	**89.79%**	**89.79%**

3.4 Their Published Results (Reference: Analytics, 2023)

Model	Accuracy	Bullying-class F1
TF-IDF + RF	84%	83%
TF-IDF + SVM	84%	85%
SBERT + RF	81%	81%
SBERT + SVM	85%	85%

Model	Accuracy	Bullying-class F1
XLM-RoBERTa	86%	86%
XLNet	86%	86%
BERT	85%	86%
RoBERTa	**87%**	**87%**

4. Key Findings Summary

- RoBERTa was the best-performing model across every experimental setting tested (our data, Esan's data, Ogunleye's data, and Ogunleye's own published results).
- Our RoBERTa outperforms the base paper (Esan et al., 2024) by a wide margin: 92.33% (our data) and 89.01% (their own data) vs. their reported 77.00%.
- Our RoBERTa outperforms Ogunleye & Dharmaraj's published RoBERTa result (87.00%) on our own dataset (92.33%).
- Ogunleye & Dharmaraj's own code implementation achieved a higher RoBERTa accuracy on our dataset (95.25%) than our own implementation (92.33%) — indicating that training/implementation details (tokenization, hyperparameters) meaningfully affect results, and represents a concrete direction for future improvement.
- Across all four experimental settings and both reference papers, RoBERTa's consistent top performance validates the choice of RoBERTa as the proposed model, while the addition of LIME provides explainability that neither Esan et al. (2024) nor Ogunleye & Dharmaraj (2023) include.